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ELEC 345 & 346

High speed communications

Wifi Architecture

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The MAC protocol

- Dynamic channel allocation.

Static model: A number of independent stations generating frames for transmission. This model assumes that once frame is generated, the station is blocked and does nothing until the frame has been usefully transmitted.

Single channel available for all users: All stations can transmit on the channel and all stations can receive from the channel. For WLANs, the time varying nature of the channel must be considered as this may result in fading of the transmitted signal due to multipath propagation effects. Hand shaking is a widely used strategy to mitigate the time varying link quality

Collision: If stations or nodes transmit at the same time, they overlap resulting in data being garbled. Wireless stations cannot detect collisions as they use a half duplex mode of operation and therefore cannot transmit and receive data at the same time. For wireless networks, collision avoidance mechanisms are employed to decrease the probability of collisions.

The MAC protocol

Time: Frame transmissions can begin at any instant, (i.e., time is continuous) or time can be divided into discrete intervals and transmission begins at the start and end of a slot. Slotted systems require wide synchronisation, which is easily achieved in centralised networks but less so in distributed networks.

Carrier sense: Stations listen to the channel before transmission of data to detect whether channel is idle or busy. For wireless networks, only nodes within radio range of each other can detect carrier on the channel.

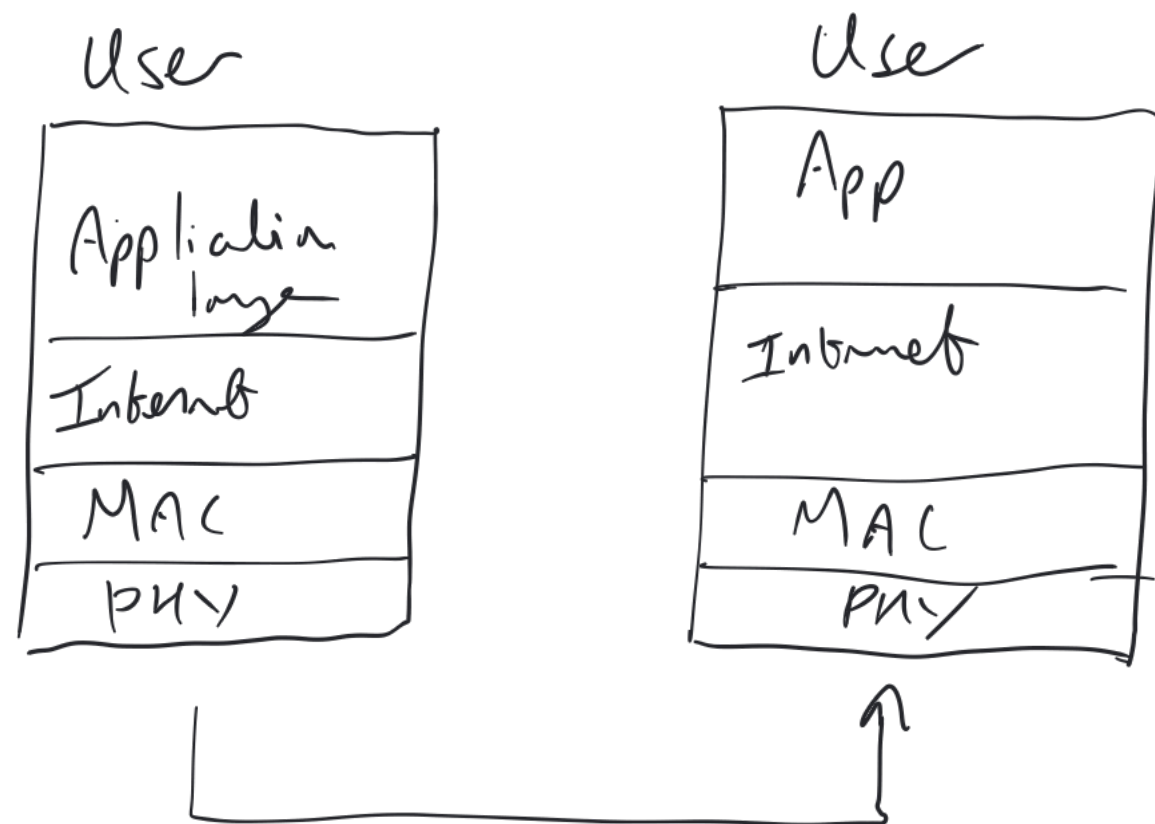
Network Architecture (wireless systems)

- Ad hoc networks have wireless terminals or nodes communicating with each other with no pre-deployed infrastructure. The nodes exchange information with each other in a distributed manner with no central administration. (TDD)
- Centralised wireless networks have a base station (BS) called an Access Point (AP) that acts as an interface between wireless and wired nodes. The nodes access the channel through the base station. The transmissions from the base station to wireless nodes are broadcast and so can be heard by all the nodes, while the transmissions to the BS is shared by all the nodes. (TDD & FDD)
- Hybrid Networks. The use of one of the previous architectures does not exclude the use of the other and many wireless LANS use one method as the primary infra-structure but allow for additional communications using the second.

The 802.11 MAC

- The basic access method is the distributed coordination function (DCF).
- Access method is a carrier sense multiple access and collision avoidance (CSMA/CA) MAC protocol.
- In addition to the DCF, the IEEE 802.11 also incorporates an alternative method known as the point coordination function (PCF) and access method that is similar to a polling system.

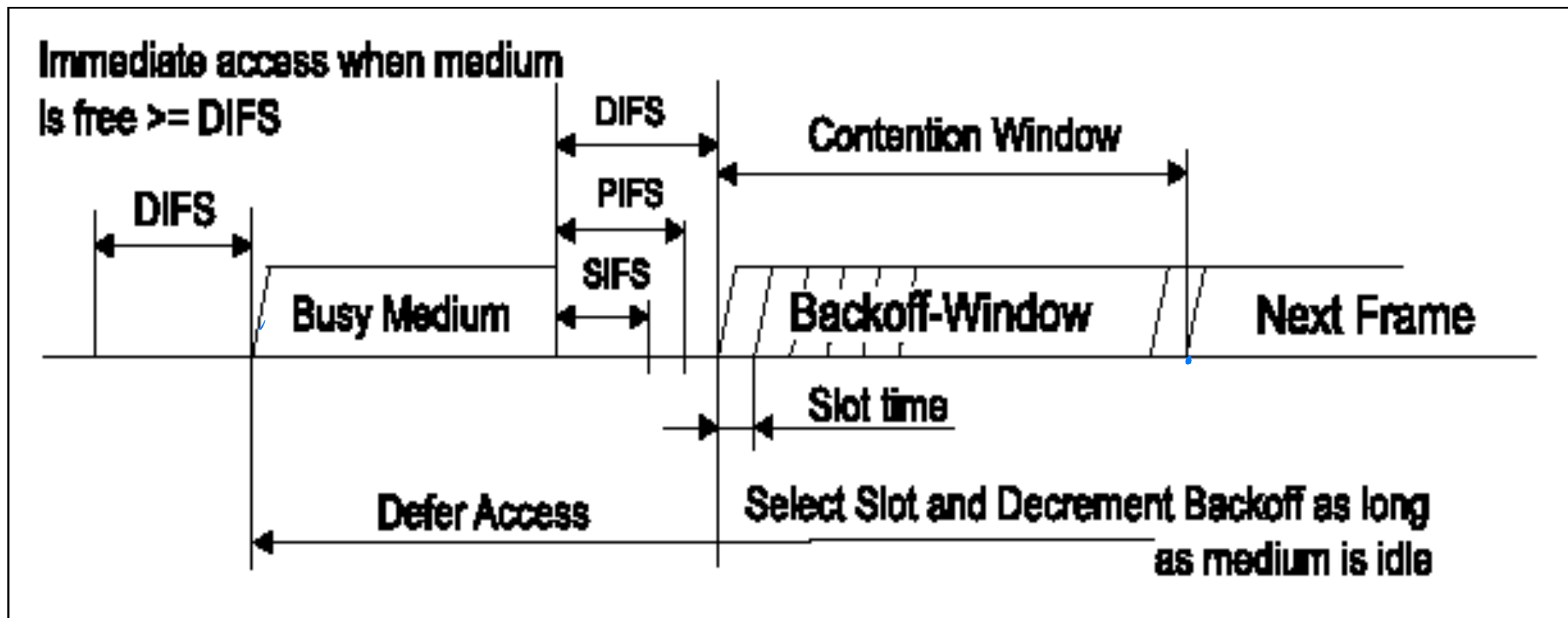
OSI
stack



Protocol stack.

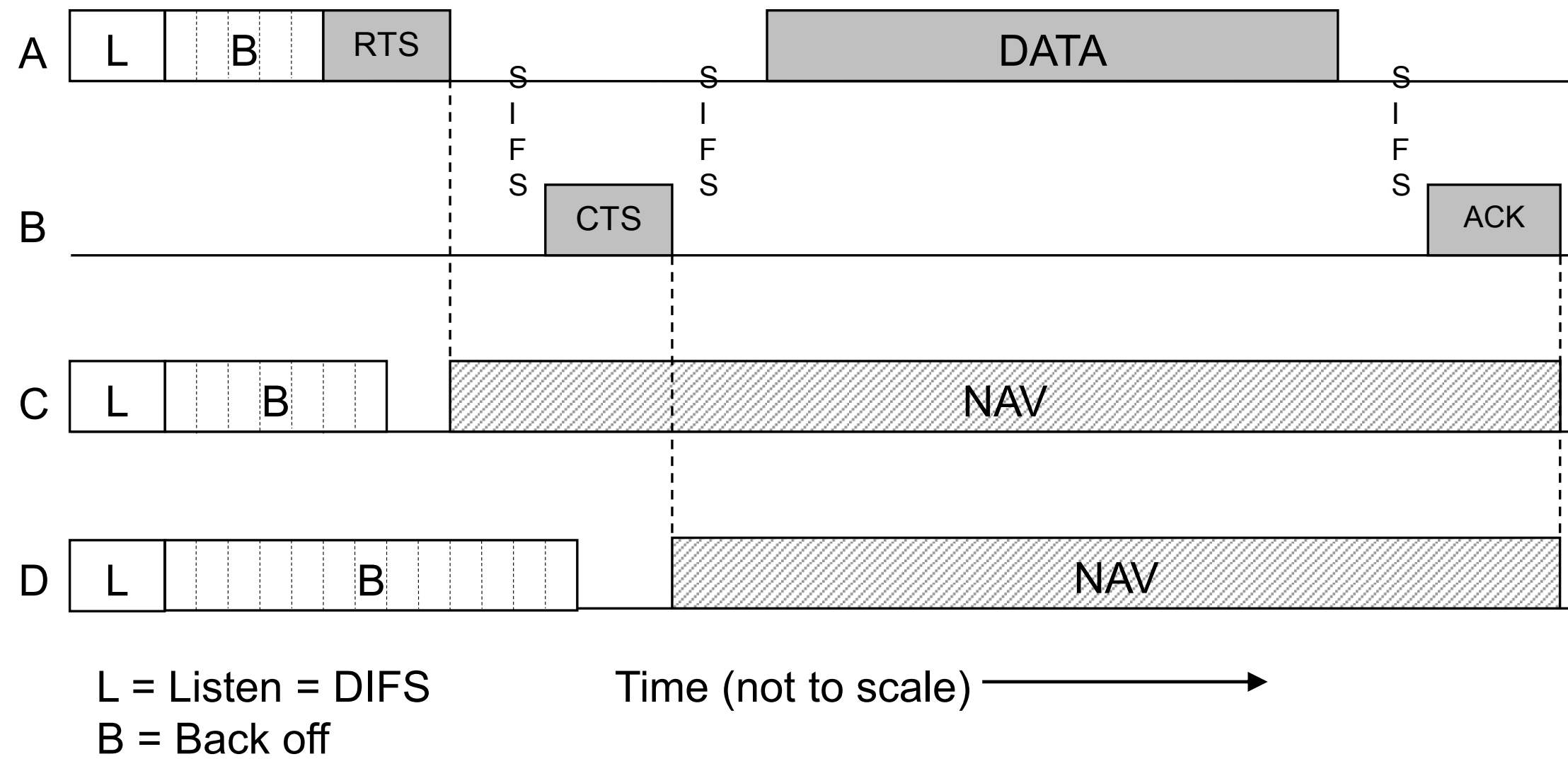
Allows the bottom part to be swapped out with little effort or discription.

Contention based channel access



In order to avoid the hidden node problem the 802.11 MAC allows nodes the option of using the earlier mentioned **RTS** and **CTS** signalling along with the network allocation vector (NAV)

Contention based channel access



Fixes hidden node problem

MAC timing

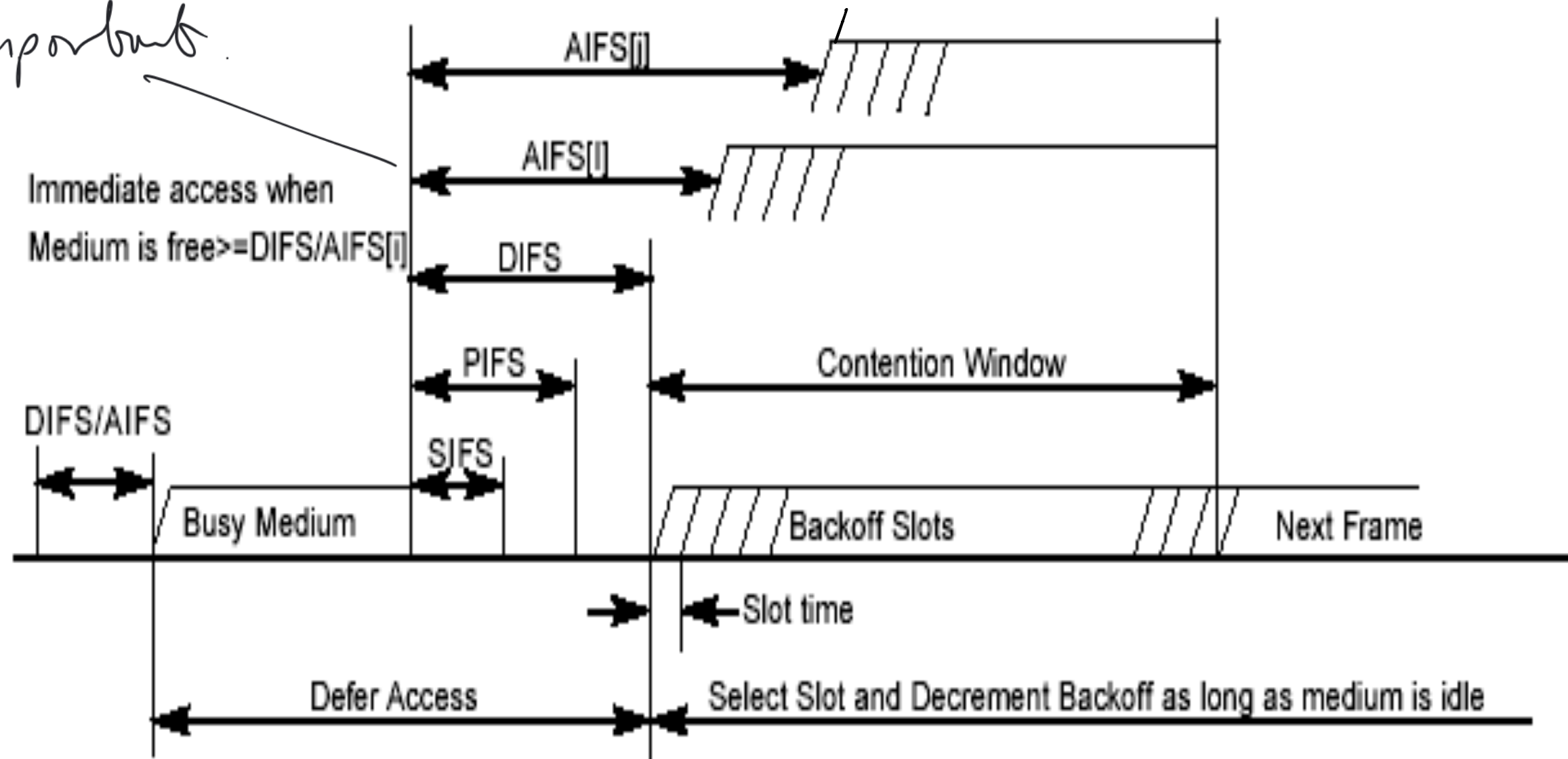
- The timing intervals used in an 802.11 system are made up of two basic blocks, the SIFS and the Slot Time.
 - The slot time is based on the time needed to listen to the channel to verify it is idle and then switch the PHY layer from receive to transmit.
 - The SIFS time is set by the time taken for the transmission over the medium, the PHY processing delay, the MAC delay and the RX-TX turnaround time. During this time other nodes should not be transmitting
- All other timing intervals are built on these
 - The PIFS time is the SIFS plus one Time Slot
 - DIFS is the SIFS plus two Time Slots
 - EIFS is the SIFS plus three Time Slots
- The particular times for each of the 802.11 systems are provided by the specific PHY

Equation 1 $\text{SIFS Time} = \text{RxRFDelay} + \text{RxPLCPDelay} + \text{MACProcessingDelay} + \text{RxRxTurnaroundTime}$

Equation 2 $\text{SlotTime} = \text{CCATime} + \text{RxTxTurnaroundTime} + \text{AirPropagationTime} + \text{MACProcessingDel}$

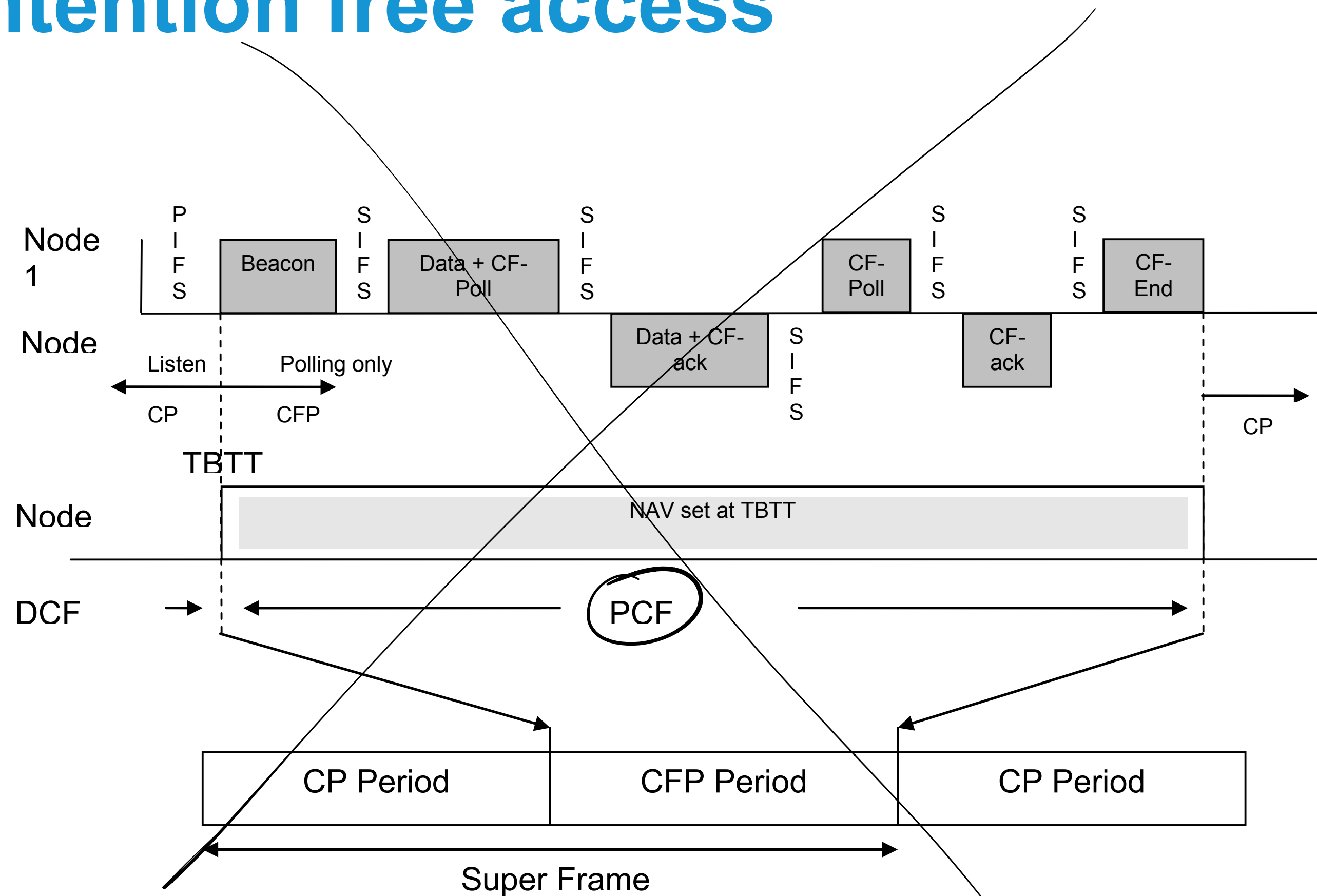
QoS ~ the enhanced DCF

Least
Most important

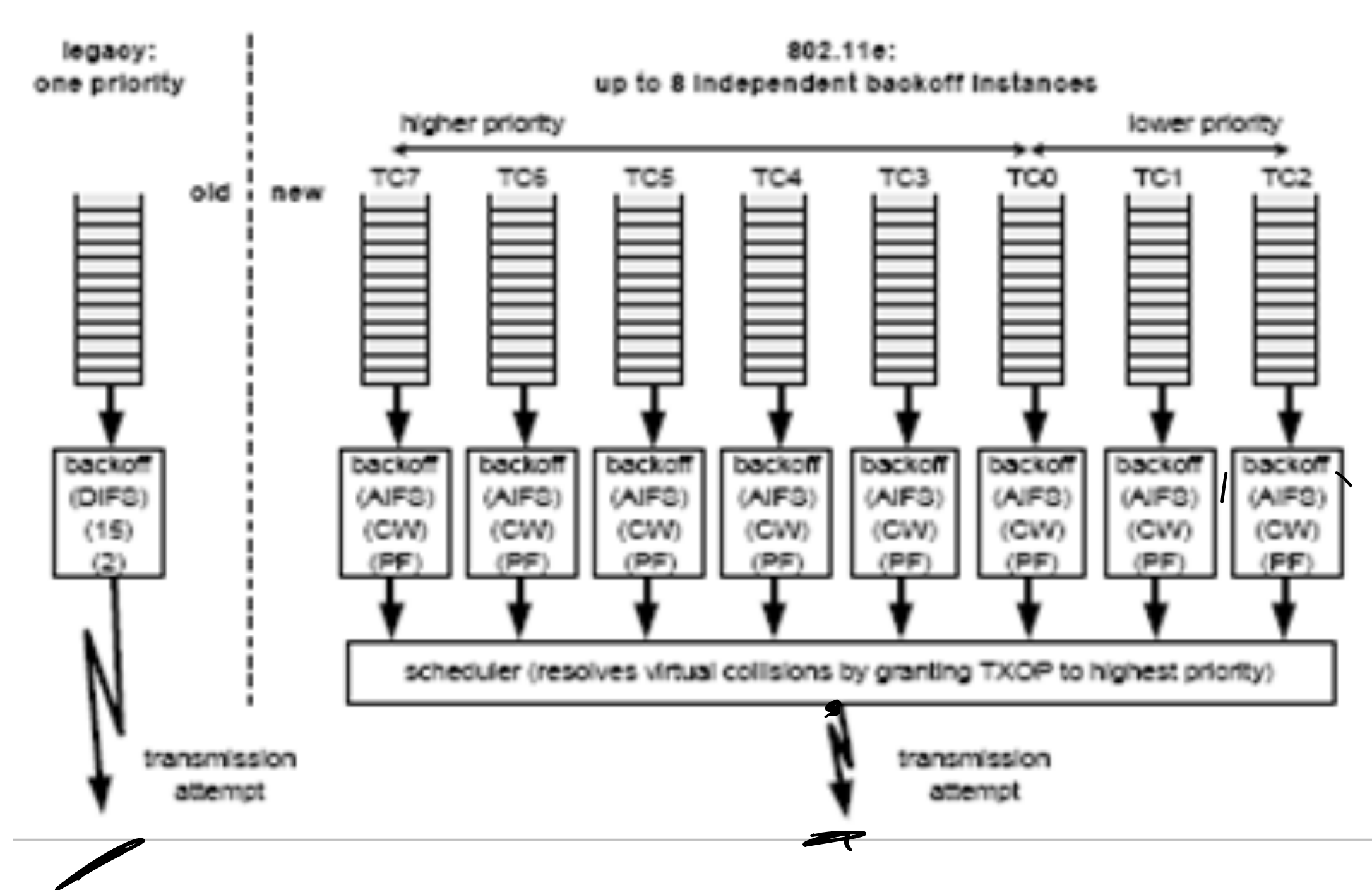


Traffic Category (802.11e)	Access Category	Designation
0	0	Best Effort
1	0	Best Effort
2	0	Best Effort
3	1	Video Probe
4	2	Video
5	2	Video
6	3	Video
7	3	Video

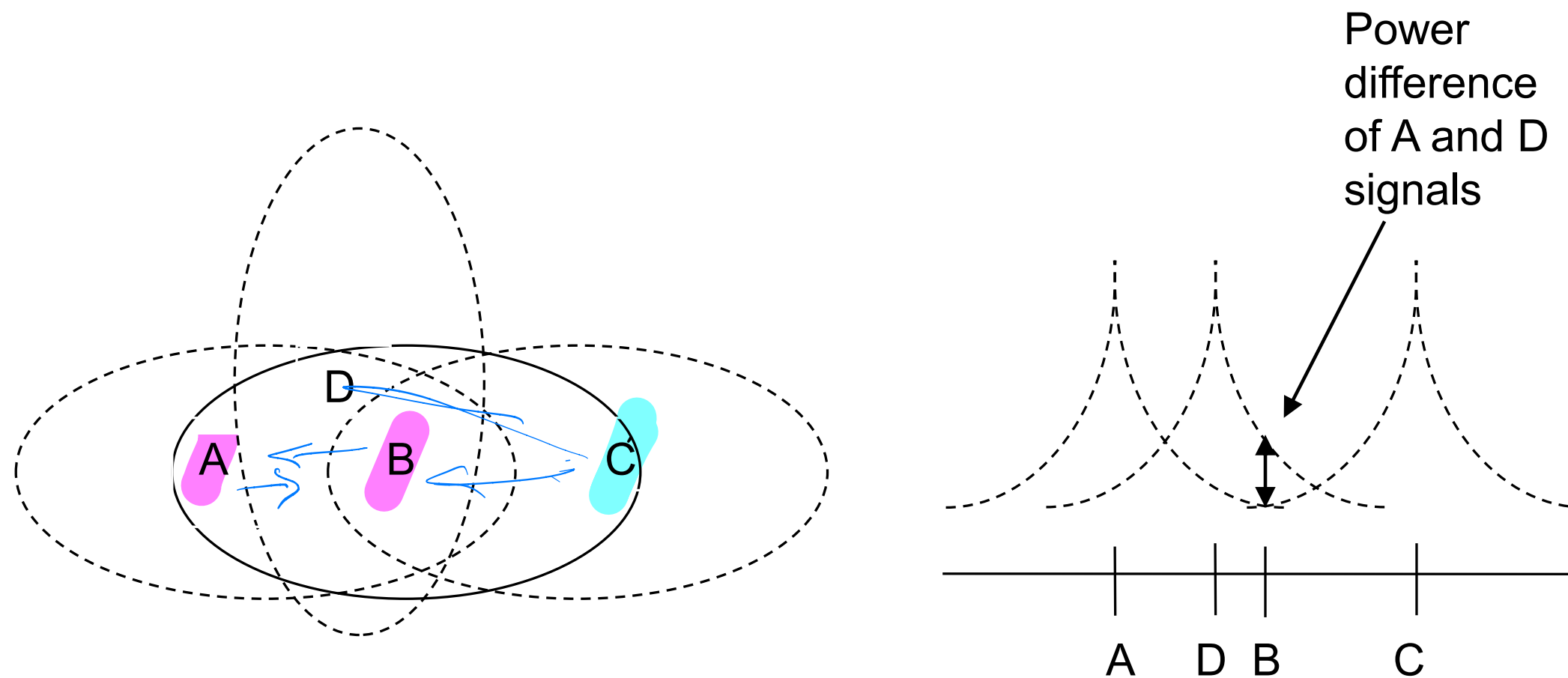
Contention free access



Legacy 802.11 and virtual queuing in 802.11e

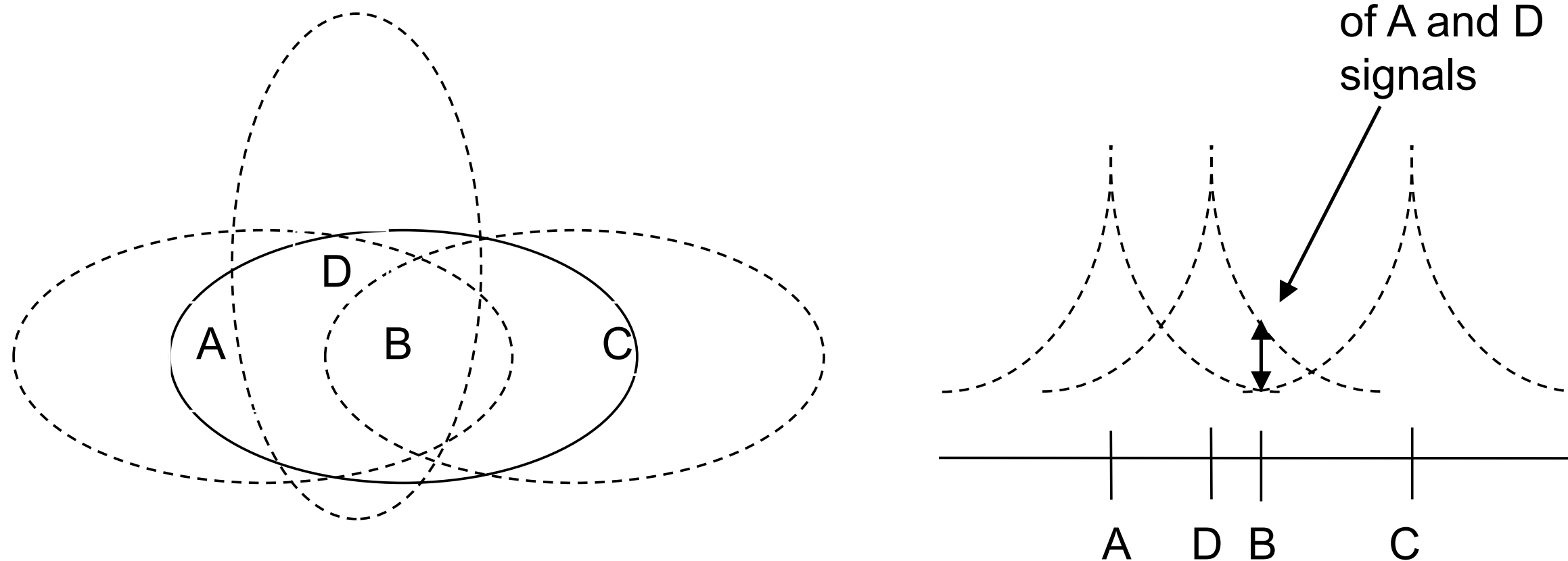


Location dependant sensing



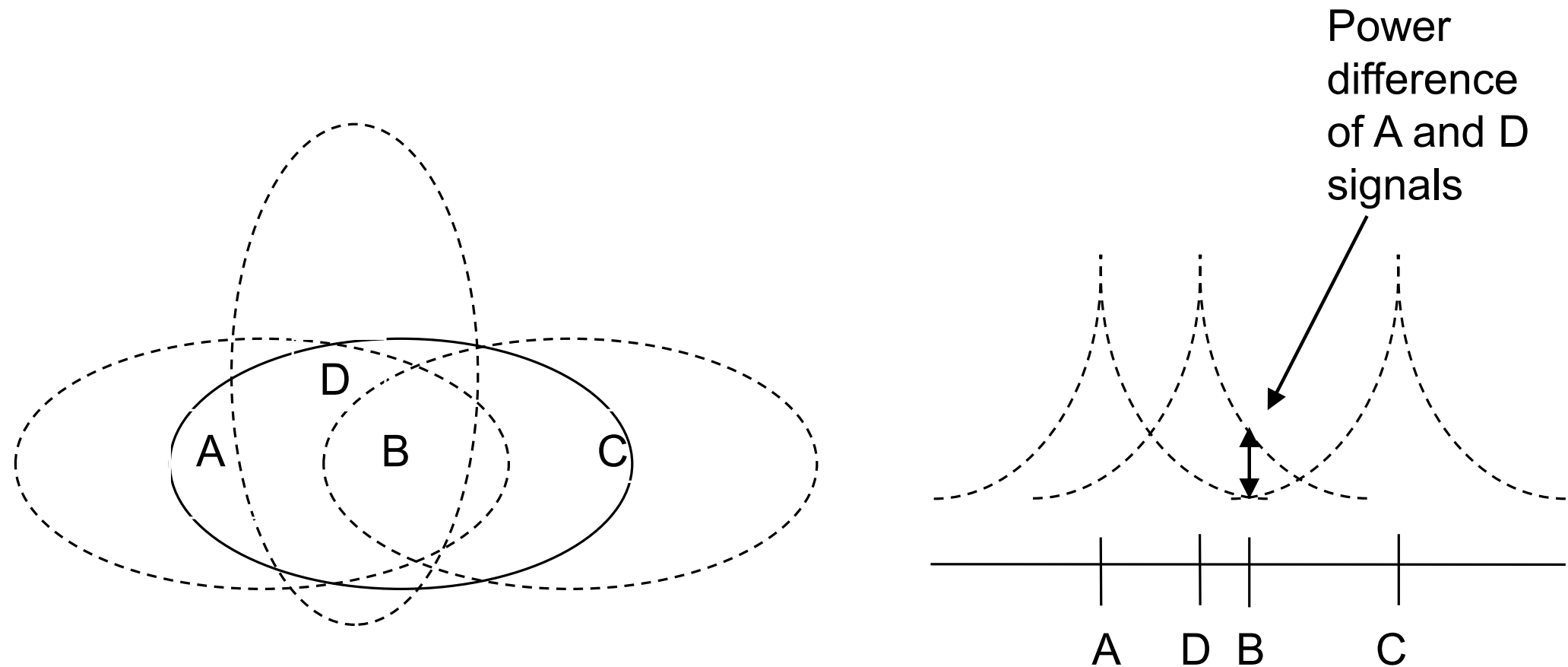
Hidden Nodes: A hidden node is one that is within the range of the intended destination but out of range of the sender. Consider the example above where node A is transmitting to node B. During this transmission when C senses the channel, it falsely thinks that the channel is idle (because C is out of range of A but in range of B). If C starts a transmission it will interfere with reception at B. In this case node C is the hidden node to node A. Hence hidden nodes cause collisions on data transmissions.

Location dependant sensing



Exposed nodes: Exposed nodes are complementary to hidden nodes. An exposed node is one that is within the range of the sender but out of range of the destination. Consider the case that node B is attempting to transmit to A. Node C can hear the transmission from B. When it senses the channel, it thinks that the channel is busy. However, any transmission by node C does not reach node A and hence does not interfere with data reception at node A. In theory, C can therefore have a parallel conversation with another terminal out of range of B and in range of C. In this case, node C is an exposed node to node B. If the exposed nodes are not minimized the bandwidth is underutilized.

Location dependant sensing



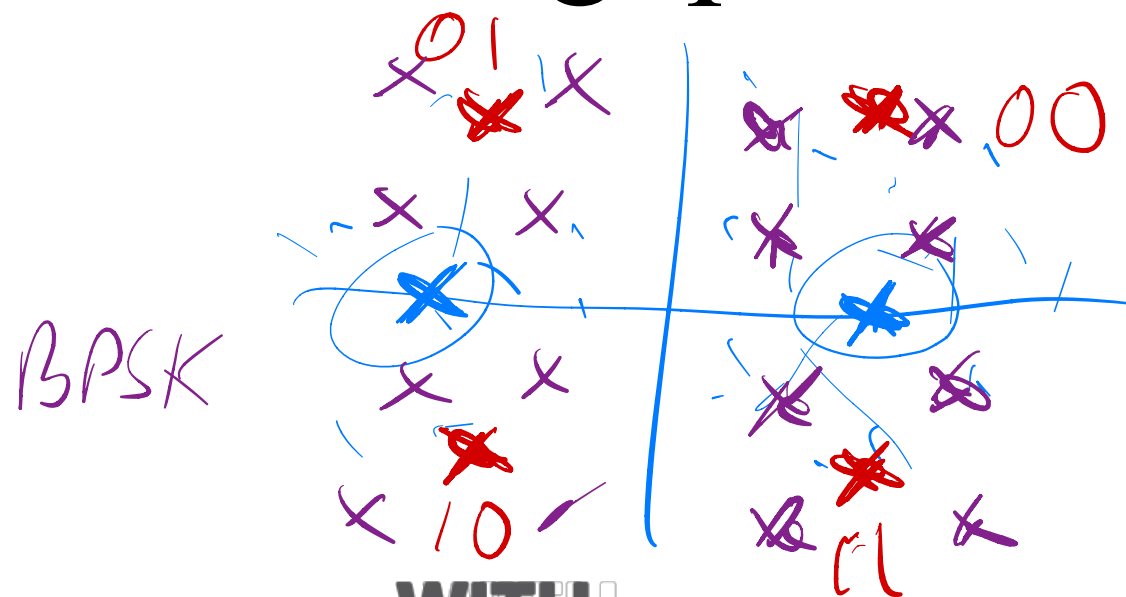
Capture: Capture is said to occur when a receiver can cleanly receive a transmission from one of two simultaneous transmissions, both within its range. In the diagram when nodes A and D transmit simultaneously to B the signal strength received from D is much higher than that from A, and D's transmission can be decoded without errors in the presence of transmission from A. Capture can improve protocol performance, but it results in unfair sharing of bandwidth with preference given to nodes closer to the BS. Wireless MAC protocols need to ensure fairness under such conditions.

Rate Adaptation

Measured performance data for 802.11a and 802.11b.

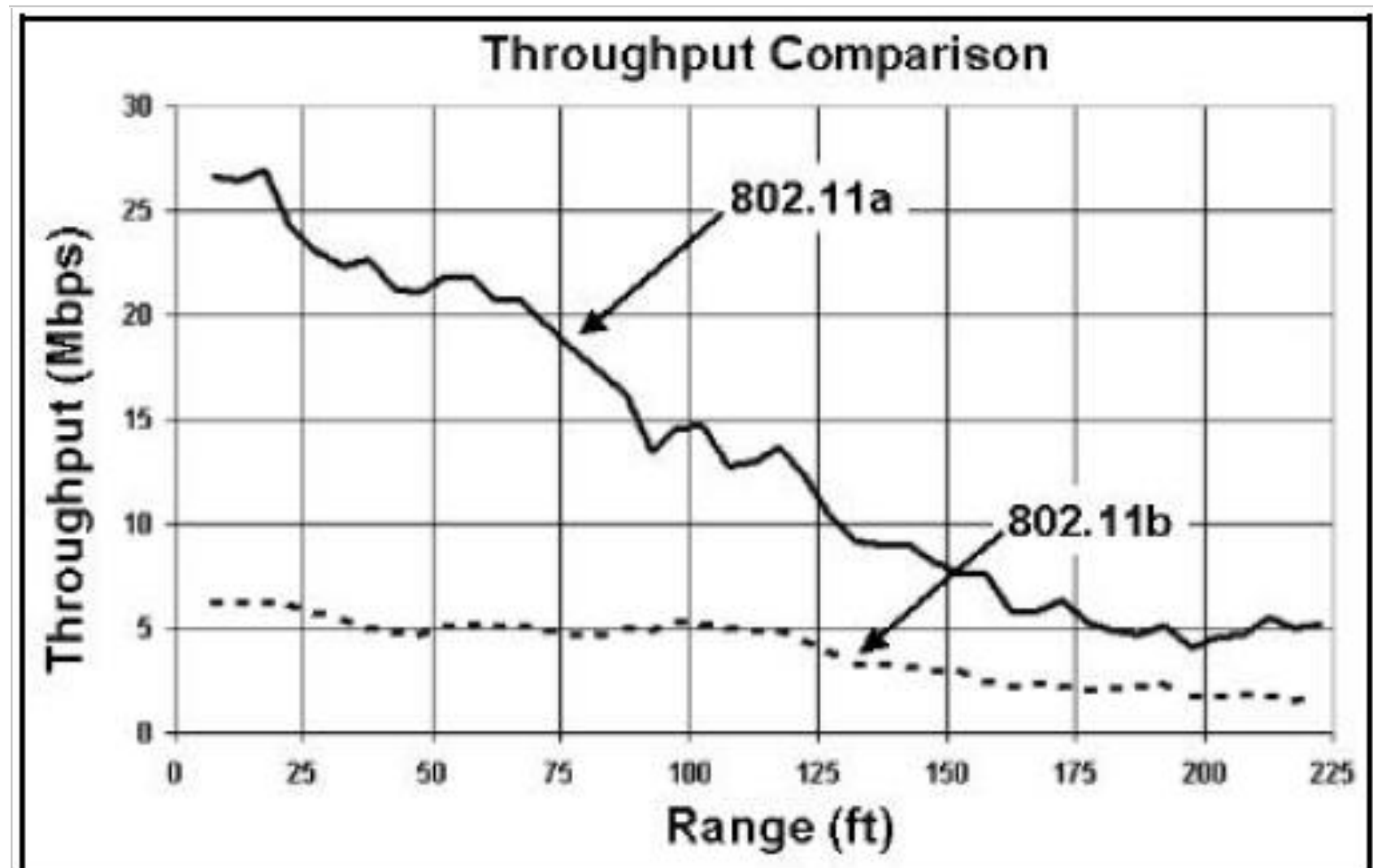
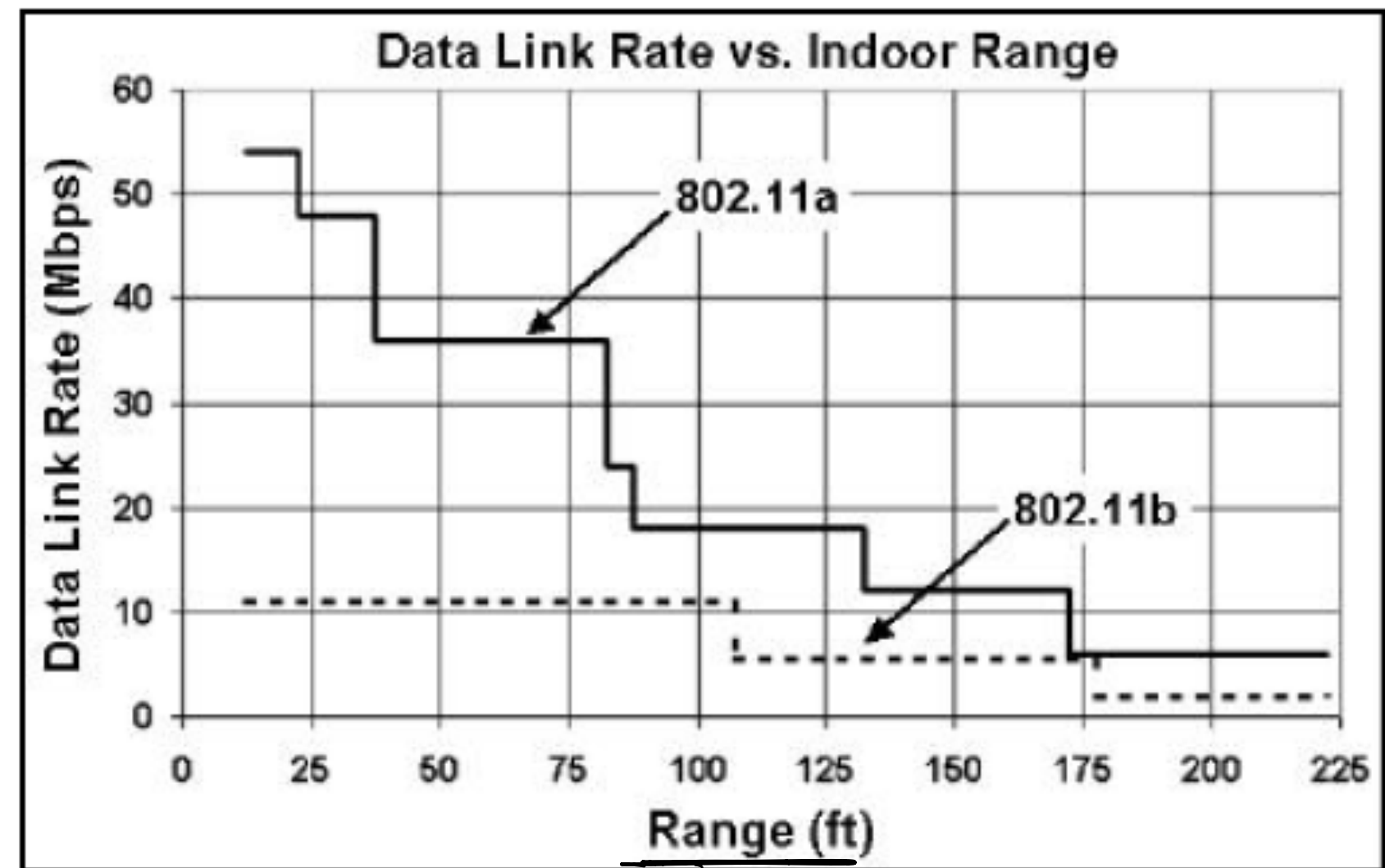
Link speed and throughput.

QPSK

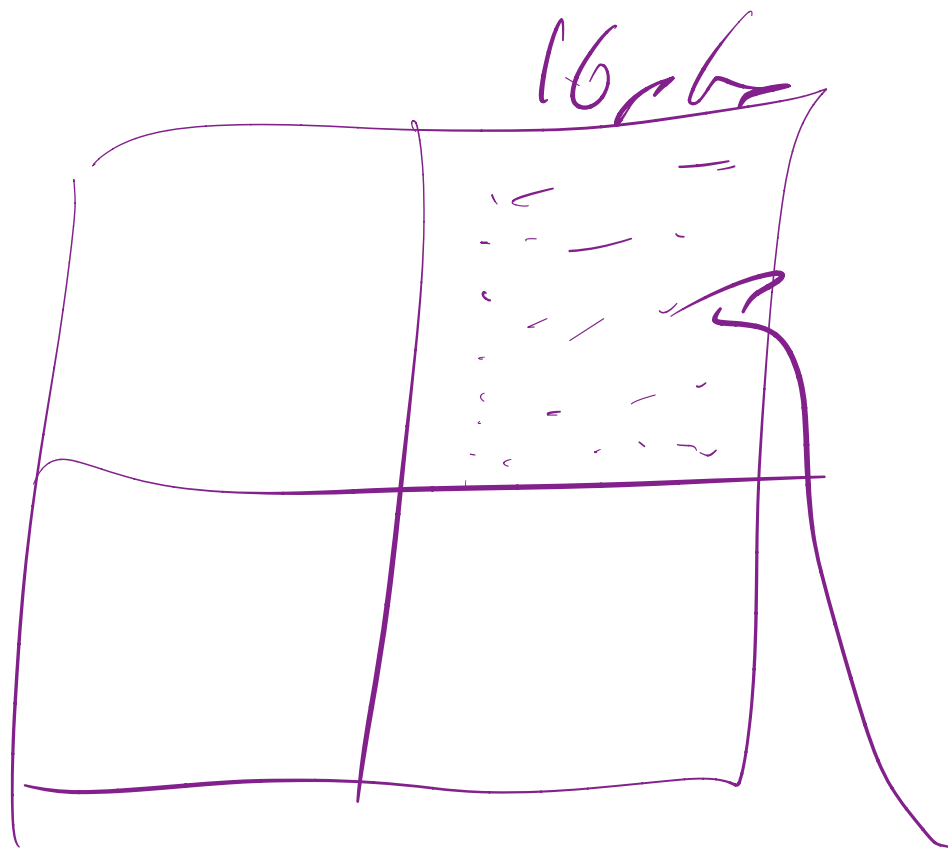
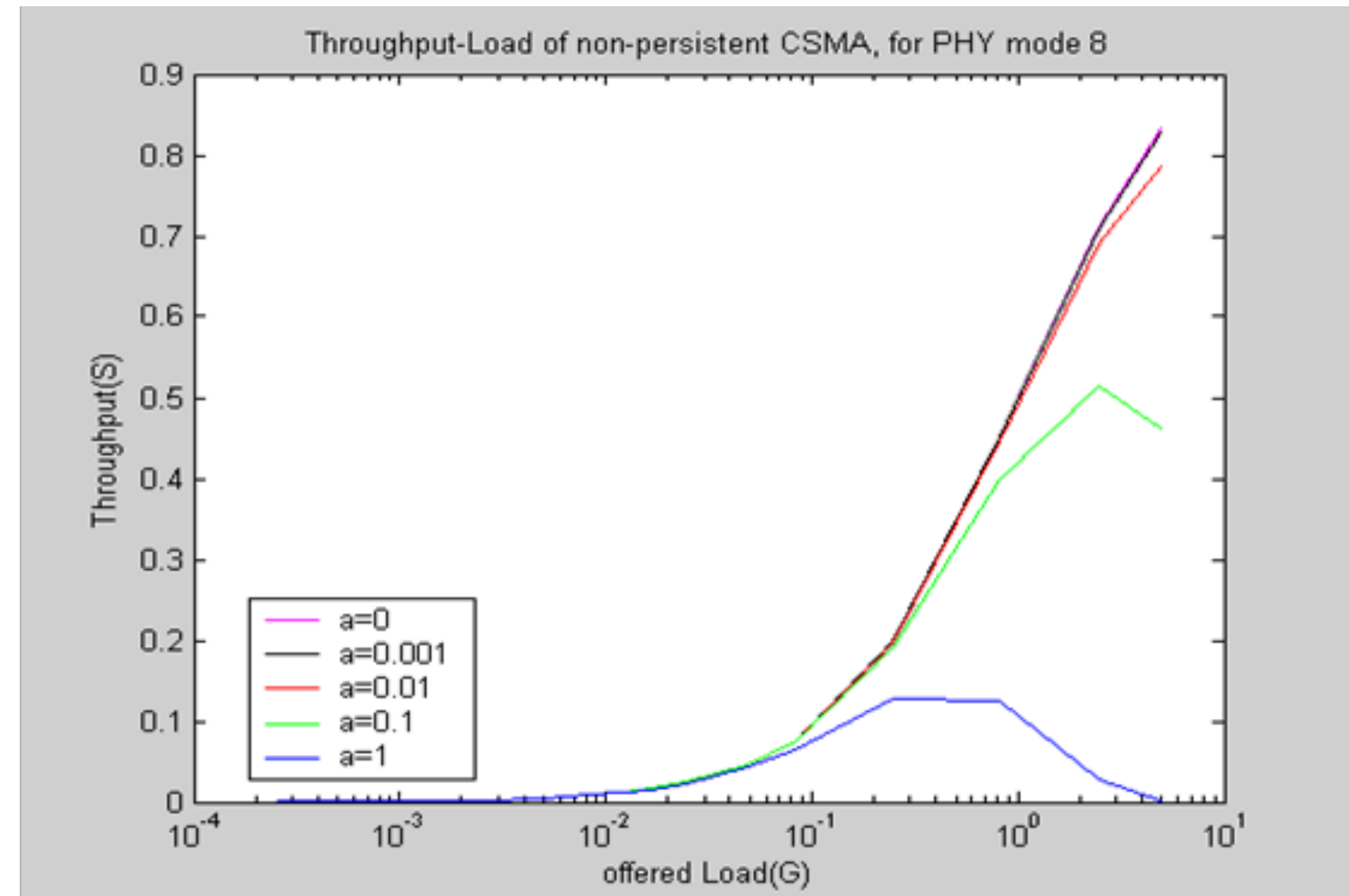
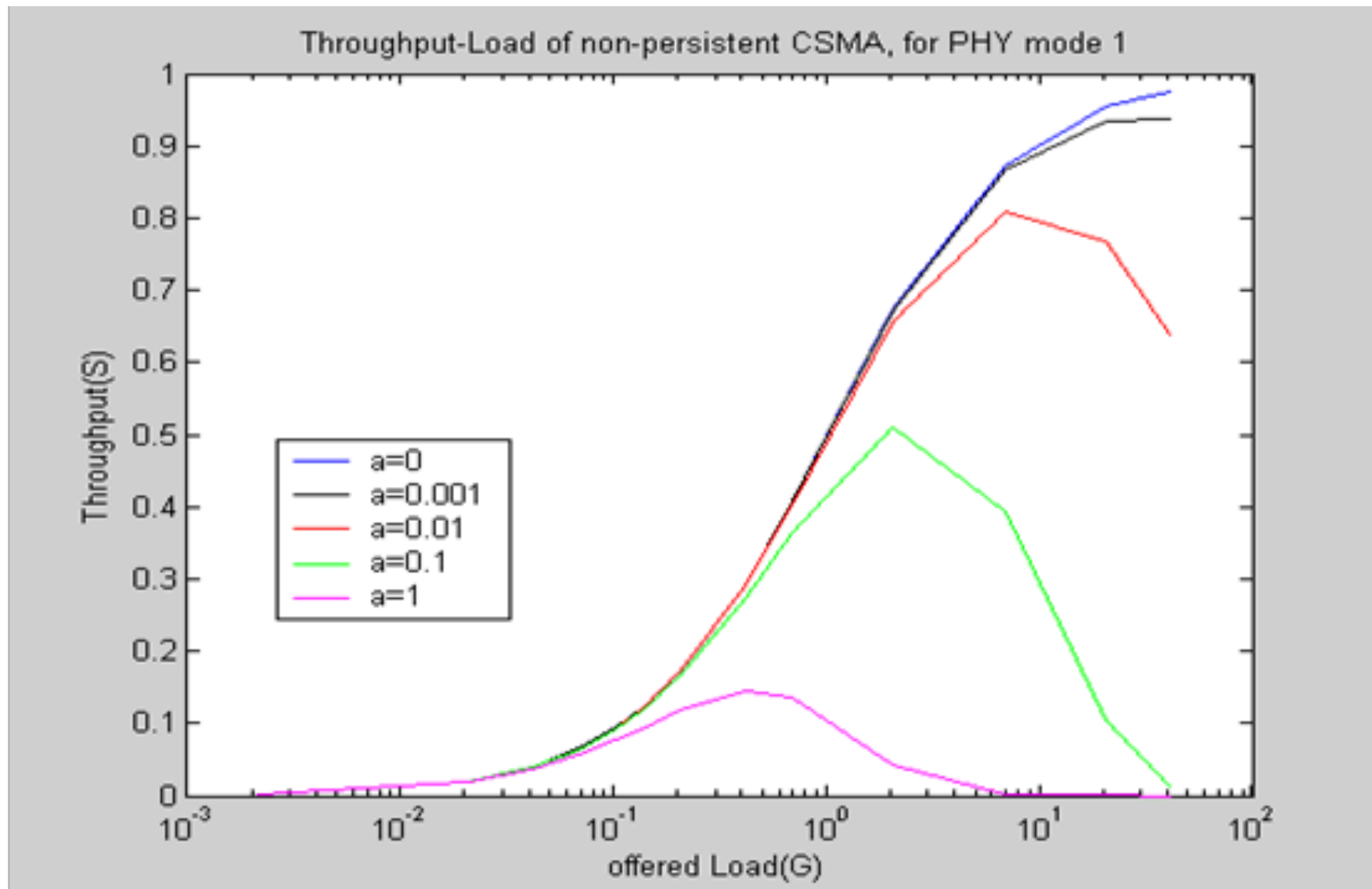


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16 QAM



Simulated throughput, 802.11a



Mode	Modulation	Code Rate	Data Rate	BpS
1	BPSK	1/2	6 Mbps	3
2	BPSK	3/4	9 Mbps	4.5
3	QPSK	1/2	12 Mbps	6
4	QPSK	3/4	18 Mbps	9
5	16-QAM	1/2	24 Mbps	12
6	16-QAM	3/4	36 Mbps	18
7	64-QAM	2/3	48 Mbps	24
8	64-QAM	3/4	54 Mbps	27



Summary

- Looked at WIFI architecture
- Understand how access is controlled DCF, ~~PCF~~
- Familiar with CSMA/CA
- Understand DIFS, SIFS and contention.
- Looked at QoS
- Looked at rate adaptation
- Looked at effect of load on throughput.